Project Code | *Kod Projek*	BITU 3973

UNIVERSITI TEKNIKAL MALAYSIA MELAKA

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

PROJEK SARJANA MUDA 1: PROPOSAL FORM

A TITLE OF PROPOSED PROJECT | *TAJUK PROJEK YANG DICADANGKAN*

ACESS - ADAPTIVE COGNITIVE & EDUCATIONAL SKILL SUPPORT PLATFORM

B DETAILS OF STUDENT | *BUTIRAN PELAJAR*

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C PROJECT INFORMATION | *MAKLUMAT PROJEK*

(i) Executive Summary of Project Proposal [Maximum 300 words]

Learning disabilities such as dyslexia, ADHD, and mild cognitive impairment affect a large group in Malaysia. In 2024, learning disabilities accounted for the highest number of registered persons with disabilities (299,128 individuals or 37.1% of total registered PWD), according to the Department of Statistics Malaysia. Traditional e-learning platforms are built for neurotypical users and often overwhelm these learners with dense text, fast-paced videos, and complex interfaces, creating significant barriers to skill development and employment.

This project proposes the development of ACESS, an accessible, role-based web platform. The system allows Educators to create structured courses with sequential lessons, transcripts, and quizzes. Learners benefit from built-in Text-to-Speech (TTS) read-aloud, adjustable font size, high-contrast themes, and a rule-based adaptive recommendation engine that automatically adjusts content difficulty based on quiz performance.

Upon course completion, the platform automatically generates verifiable PDF certificates with unique reference codes. A progress tracking module and educator analytics dashboard further support personalized learning and timely intervention.

ACESS aims to deliver an inclusive digital learning environment that improves learning outcomes, reduces dropout rates, equips learners with job-ready skills, and provides employers and disability support organizations with credible evidence of competencies.

(ii) Detailed Proposal of the Project

(a) Introduction *(Project Background and Problem Statements)*

Learning disabilities such as dyslexia, ADHD, and mild cognitive impairment affect many people around the world. Globally, approximately 15 to 20 percent of individuals experience some form of learning or thinking difference. In Malaysia, a total of 805,509 persons with disabilities were registered in 2024, representing 2.4 percent of the population. Among them, learning disabilities recorded the highest number with 299,128 individuals, or 37.1 percent of all registered cases.

These conditions do not indicate lower intelligence. They simply reflect different ways the brain processes information. However, most existing e-learning platforms are designed for neurotypical learners and provide little to no support for cognitive, sensory, or attentional needs. As education and vocational skill training continue to shift online, learners with disabilities often feel overwhelmed by dense text, fast-paced videos, and complex interfaces. Without suitable accommodations such as adjustable text size, read-aloud features, simpler navigation, and flexible pacing, many experience frustration, disengagement, and high dropout rates.

This project seeks to address these challenges by developing ACESS - Adaptive Cognitive & Educational Skill Support Platform. The system will offer a complete, accessible digital learning journey from user registration to certificate generation. It includes adaptive content recommendations, built-in accessibility tools, progress tracking, and dedicated dashboards for educators and administrators.

Problem Statements

1. Most existing e-learning platforms are not designed for learners with disabilities. They use formats that overwhelm cognitive and sensory processing, leading to high dropout rates.
2. Learners with disabilities lack platforms that adapt to their individual learning pace and present content in accessible, easy-to-process formats.
3. Employers and support organisations have no reliable way to verify the skills gained by learners with disabilities due to the absence of proper progress tracking and accessible certification.
4. Educators and mentors do not have a centralised platform to monitor learner progress, identify those who need extra support, and manage accessible learning materials effectively.

(b) Objectives of the Project

This project aims to achieve the following objectives:

1. To analyze the accessibility requirements, learning challenges, and content delivery needs of learners with dyslexia, ADHD, and mild cognitive impairment through literature review and comparative study of existing e-learning platforms and assistive technology standards.

2. To develop a fully functional accessible web-based e-learning platform that incorporates rule-based adaptive content delivery, built-in Text-to-Speech (TTS) support, adjustable accessibility display settings, structured lesson and quiz management, and role-based user management for learners, educators, and administrators.

3. To implement a certificate generation module and an educator analytics dashboard that provide verifiable skill completion records and enable data-driven monitoring of individual learner progress.

4. To evaluate the functional correctness, usability, and accessibility of the developed platform through structured user acceptance testing (UAT), measuring task completion rates and user satisfaction using the System Usability Scale (SUS).

(c) Scope of the Project

System Modules:

The project will cover all the following modules to develop a comprehensive system.

1. Authentication and User Management Module
   - Handles secure registration with email verification, login, logout, and password reset.
   - Implements Role-Based Access Control (RBAC) for Learner, Educator, and Administrator roles.
   - Allows administrators to create, edit, activate/deactivate user accounts and assign roles.
   - Includes learner onboarding profile for declaring disability type, accessibility preferences, and learning goals.

2. Course and Lesson Management Module
   - Enables educators to create, edit, and publish structured courses with sequential lessons, titles, descriptions, and skill tags.
   - Supports text content, embedded videos with transcripts, and prerequisite sequencing for scaffolded learning.
   - Provides learners with a visual progress bar and module completion checklist.

3. Adaptive Content Recommendation Module
   - Uses rule-based logic to adjust difficulty based on quiz scores (revision below 60%, standard 60–80%, advanced above 80%).
   - Displays personalized “Recommended Next” cards on the learner dashboard.
   - Logs all recommendations for educator review.

4. Quiz and Assessment Module
   - Allows educators to create multiple-choice and scenario-based quizzes with time limits and attempt controls.
   - Presents questions one at a time in a clean, distraction-free interface with large text and progress indicators.
   - Provides immediate feedback and feeds scores into the adaptive recommendation system.

5. Progress Tracking Module
   - Records lesson views, quiz attempts, scores, and timestamps for each learner.
   - Displays a simple personal dashboard for learners showing completion percentage and next steps.
   - Enables educators to view individual progress and identify at-risk learners.
   - Provides administrators with platform-wide completion and engagement statistics.

6. Accessibility Settings Module
    - Allows learners to customize font size, display theme (light/dark/high contrast), and line spacing.
    - Integrates a one-click Text-to-Speech (TTS) button for lesson content.
    - Provides toggleable transcripts for video content.
    - Saves all preferences to the user account for automatic restoration on login.

7. Certificate Generation Module
    - Automatically generates downloadable PDF certificates upon course completion with learner name, course title, date, and unique reference code.
    - Provides learners with a dedicated page to view and download all earned certificates.
    - Allows administrators to view, manage, revoke, or reissue certificates.

8. Educator Analytics Dashboard Module
    - Displays course enrollment, average quiz scores, completion rates, and at-risk learners.
    - Visualizes quiz score distributions using bar charts.
    - Supports filtering by course, date range, and disability type.
    - Allows export of progress reports in CSV format.

Target Users

1. Learners (Users with Learning Disabilities)
    - Self-register and complete onboarding profile with disability type and accessibility preferences.
    - Learn at their own pace using TTS, transcripts, and adjustable display settings.
    - Complete quizzes, receive immediate feedback, and follow adaptive recommendations.
    - Track progress and download certificates for job applications or program reporting.

2. Educators (Content Creators and Mentors)
    - Create structured courses with lessons, transcripts, and quizzes.
    - Monitor individual and group progress through the analytics dashboard.
    - Identify at-risk learners and adjust content based on performance data.
    - Review adaptive recommendation logs to ensure appropriate progression.

3. Administrators (System Managers)
    - Manage all user accounts, roles, and registrations.
    - Review and approve courses before publication.
    - Monitor platform-wide statistics and generate reports for stakeholders.
    - Oversee certificate issuance and platform usage.

(d) Expected Outcome/ Proposed Solution

The proposed solution is the development of a centralized, role-based accessible e-learning web application. The platform is specifically designed to support learners with dyslexia, ADHD, and mild cognitive impairment, along with their educators and administrators, by providing an inclusive end-to-end learning experience from registration to certificate issuance.

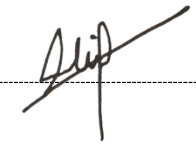
Upon successful completion, the project is expected to deliver the following key outcomes:

- A fully functional accessible web platform that provides learners with dyslexia, ADHD, and mild cognitive impairment a disability-friendly learning environment, including built-in Text-to-Speech (TTS), adjustable font size, display themes, line spacing, and lesson transcripts.
- An adaptive content recommendation system that automatically adjusts lesson difficulty based on quiz performance (easier revision for scores below 60%, standard progression for 60–80%, and advanced content for scores above 80%), ensuring learners study at an appropriate pace.
- An automatic certificate generation module that produces downloadable PDF certificates with unique reference numbers upon successful course completion, allowing learners to present verifiable proof of their skills to employers and support organizations.
- A user-friendly educator analytics dashboard that displays course progress, quiz score distributions, completion rates, and at-risk learners, enabling educators to monitor progress and make timely adjustments to content or support.
- A secure, role-based system with clear separation of access and features for Learners, Educators, and Administrators, enforced through Laravel's Role-Based Access Control (RBAC).

D REFERENCES | *RUJUKAN*

State your references (Minimum 10 references)

1	*https://www.dosm.gov.my/portal-main/release-content/person-with-disability-statistics-malaysia-2024*
2	*https://www.singhealth.com.sg/symptoms-treatments/mild-cognitive-impairment*
3	*https://pmc.ncbi.nlm.nih.gov/articles/PMC9554924/*
4	*https://crowncounseling.com/statistics/learning-disabilities/*
5	*https://www.wholeschoolsend.org.uk/page/cognition-and-learning*
6	*https://www.youtube.com/watch?v=fm3SDn7pUGk*
7	*https://www.talentlms.com/*
8	*https://360learning.com/*
9	*https://nextjs.org/*
10	*https://firebase.google.com/*

E	**DECLARATION BY STUDENT** | *AKUAN PELAJAR*
(i)	**Date:** 4/1/2026 
E	**RECOMMENDED BY SUPERVISOR** *PERAKUAN OLEH PENYELIA*
	Recommended ☐
	Not Recommended ☐
	Comments:
	Supervisor's Name:
	Signature & Stamp:
	Date: